EE1007
Introduction to Computer Science Laboratory

Lab 12
Sequential-Access Files

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Lab 12.1
Find Failed Students

- In this problem, you need to write a program which can find out all the failed students from a given score file.
- In the beginning, the program will take a file name as input, and the file contains the score of all students.
- The range of each student's scores is [0,100].
- Then the program will show two numbers, first number is the number of students in the class, and the second number is the number of failed students in the class. Note that "failed student" means a student whose score is less than 60.

Lab 12.1
Find failed student

- If the file name entered by user does not exist, the program will show an error message "File could not be opened." and then exit the program.
- Because of the restrictions on OJ, the object/function **"cerr"** and **"exit(1)"** are not allowed to use in this program
  - The above object/function will let OJ show the error message "runtime error", so please avoid using these instructions.
  - cerr << "File could not be opened. " << endl;
    exit(1); (X)
  - cout << "File could not be opened. " << endl;
    exit(0); (O)

Lab 12.1
Find failed student

- The screen dialog should appear as follows:

```
Please enter the input file name:
123.txt
File could not be opened.

Process returned 1 (0x1)   execution time : 3.211 s
Press any key to continue.
```

```
Please enter the input file name:
1_1.txt
8
4

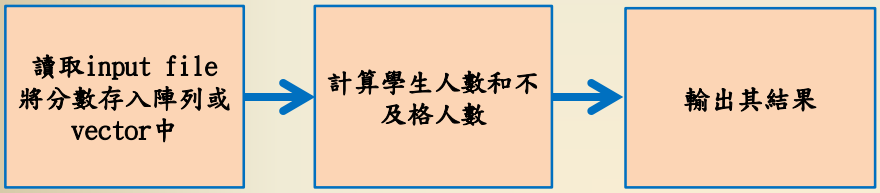
Process returned 0 (0x0)   execution time : 3.702 s
Press any key to continue.
```

1_1.txt

```
12
50
64
84
95
82
10
0
```

Lab 12.1
Find failed student

◆思考方向：



Lab 12.1
Find failed student

◆Step. 1

```
cout << "Please enter the input file name: " << endl;
cin >> infilename;

infile.open(infilename, ios::in);

if(!infile)
{
    cout<<"File could not be opened. "<< endl;
    exit(0);
}
else
{
    while (infile >> data)
    {
        num.push_back(data);
    }
    infile.close();
}
```

Lab 12.1
Find failed student

◆Step. 2

```
for(int i = 0; i < num.size(); i++)
{
    if(num[i] < 60)
        temp++;
}
cout << num.size() << endl << temp << endl;
```

學生人數　　不及格學生人數

Lab 12.2
Christmas Gift Exchange

- Since Christmas is coming soon, NCU EE class decided to play Christmas gift exchange game. For efficiency, we write an automated program which can help us to figure out the gift you obtain.
- In the beginning, the program will take a file name as a input, where the file contains all the students' presents and their number.
- Each presents in the file will contain no more than 20 characters, and all the presents only consist of numbers and letters. The numbers of present will not exceed 50.
- Then, the program will take a gift number as input and save the gift you obtain to a new file.
- Finally, you will get an output file which show the message **"Congratulations! Your Christmas present is XXX".**

Lab 12.2
Christmas Gift Exchange

- If the file name user enters does not exist, the program will show an error message "File could not be opened." and then ask the user to enter again.
- Notice that if the input file name is **"2_1.txt",** then the corresponding output file name should be **"2_1out.txt". To realize this function, t**he template of this lab is given as "Lab12_2_sample.cpp".
- Moreover, after the user chooses a gift number, the program will show the message **"Okay! Your Christmas present has been saved to XXXout.txt".** where **XXX means the input file name.**
- Notice that if the user enter a gift number which exceeds the maximum number of gifts in the file, the program will show the message "Invalid input! Please enter again: ". and ask the user to enter a number again.

Lab 12.2
Christmas Gift Exchange

- The screen dialog should appear as follows:

```
Please enter the input file name:
123.txt
File could not be opened.
Please enter the input file name:
2_1.txt
Please choose a lucky number :
6
Invalid input! Please enter again:
5
Okay! Your Christmas present has been saved to 2_1out.txt

Process returned 0 (0x0)   execution time : 18.682 s
Press any key to continue.
```

2_1.txt

No.	Present
1	Iphone13
2	PS4
3	Switch
4	Airpods
5	sneakers

Lab 12.2
Christmas Gift Exchange

◆The output file should appear as follows:

2_1.txt

Input file

No.	Present
1	Iphone13
2	PS4
3	Switch
4	Airpods
5	sneakers

A user enter 5 as the input

show on the screen: Okay! Your Christmas present has been saved to 2_1out.txt

output file: Congratulations! Your Christmas present is sneakers — 2_1out.txt

Lab 12.2
Christmas Gift Exchange

◆思考方向：

讀取input file 將禮物和對應數字存入陣列或vector中 → 根據user輸入之數字找到對應的禮物 → 將得到禮物的結果存入output file

Lab 12.2
Christmas Gift Exchange

◆Step. 1

```
A:  cout << "Please enter the input file name: " << endl;
    cin >> infilename;
    outfilename = infilename.substr(0,infilename.length()-4)+"out.txt";

    infile.open(infilename, ios::in);

    if(!infile)
    {
        cout<<"File could not be opened. "<< endl;
        goto A;
    }
    else
    {
        infile.seekg(15,ios::beg);
        while (infile >> data >> present)
        {
            gift.push_back(present);
        }
        infile.close();
    }
```

防呆機制

使用seekg讓程式不要讀到檔案第一行內容

將禮物和數字存入vector

Lab 12.2
Christmas Gift Exchange

◆Step. 2

```
cout << "Please choose a lucky number :" << endl;
cin >> num;
while(num > gift.size())
{
    cout << "Invalid input! Please enter again: " << endl;
    cin >> num;
}
cout << "Okay! Your Christmas present has been saved to " << outfilename << endl;
outfile.open(outfilename, ios::out);
outfile << "Congratulations! Your Christmas present is " << gift[num-1] << endl;
outfile.close();
```

將結果存入XXXout.txt中

Lab 12.3

NCU Theater Reservation System(IV)

- As the member number grows up, the system manager want to use files to maintain the member data.
- In this Lab, you are asked to add a function which can load data of members from a given file.
- Once a user enters 7 in main menu, the program will load member information from the stored file.
- First, the program will print the message below.
  - "***Load member information***"
- Then, the program will take a file name as input, and the file contains all the members' name and their password.
- If the file does not exist, the program will show an error message "File could not be opened." and return to the main menu.
- If the file exists, the program will read the file line-by-line and **append** the member information from the file.
- Finally, the program will show the message "The file xxx has been loaded..." (*xxx* is the input file's name).

Lab 12.3

NCU Theater Reservation System(IV)

- In this lab, we assume all member data from the file will be different from the registered account. Therefore, you do not need to consider the duplications.
- We also assume the system will only load one file.
- Based on the feedback from the system user, we have to make a minor adjustment
  - After log-in the NCU member system, the log-out function is designed when user input 7.
  - To avoid the conflict of the file loading selection, now the system will trigger the log-out function when the user input 5 again.
  - Specifically, the message on the original menu **"Please type 5 to member center"** will be changed to **"Please type 5 to log-out" .**

Lab 12.3
NCU Theater Reservation System(IV)

- After updating, the main menu after you log-in is shown as follows:

```
*** Welcome to NCU theater reservation system ***
***Welcome back Andy***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 6 to show member
Please type 7 to load member information
Please type 8 to exit the system
Please type 9 to log-out
Enter your selection Here:
```

Before

```
*** Welcome to NCU theater reservation system ***
***Welcome back Andy***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to log-out
Please type 6 to show member
Please type 7 to load member information
Please type 8 to exit the system
Enter your selection Here:
```

After

Lab 12.3
NCU Theater Reservation System(IV)

- Requirement:
  - If the user enters a wrong input, show the message **"Invalid input, please enter your selection again:"**
- Notice that the program should work at normal mode when there is no one enter the member system.

Lab 12.3
NCU Theater Reservation System(IV)

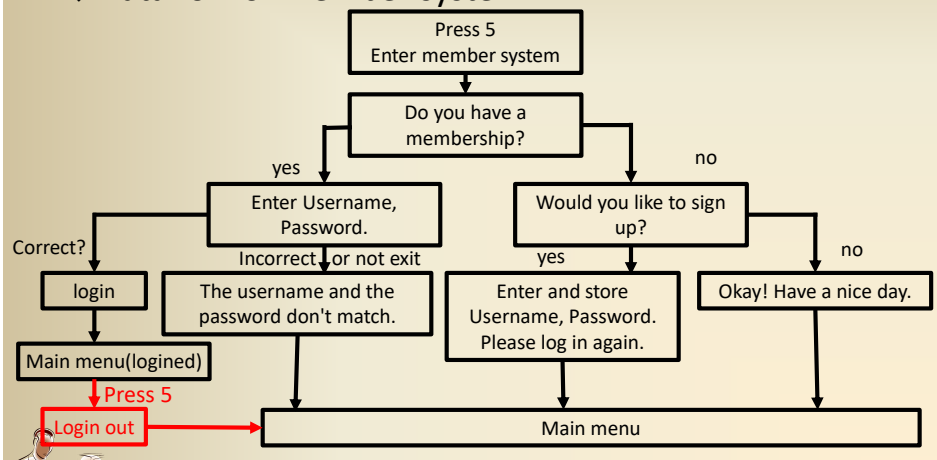
◆Quick summary

➢ The main menu of your system is shown as follows:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to load member information
Please type 8 to exit the system
Enter your selection Here:
```


Lab 12.3
NCU Theater Reservation System(IV)

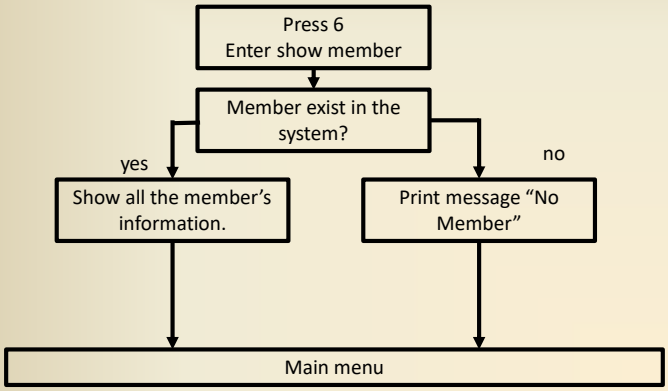
◆Dataflow of member system



Lab 12.3

NCU Theater Reservation System(IV)

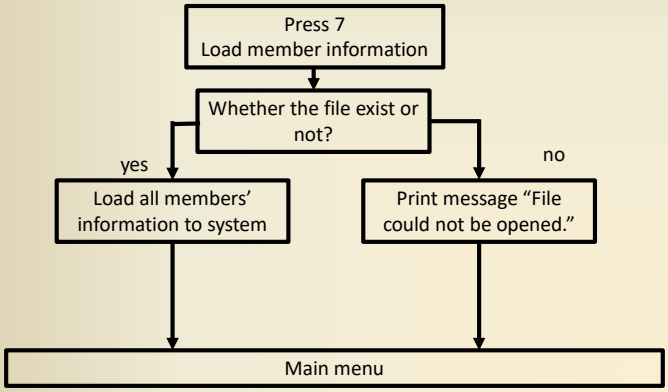
◆Dataflow of show member



Lab 12.3

NCU Theater Reservation System(IV)

◆Dataflow of load member



Lab 12.3

NCU Theater Reservation System(IV)

◆The screen dialog should appear as follows:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to load member information
Please type 8 to exit the system
Enter your selection Here: 7

***Load member information***
Please enter the member file name: 123.txt
File could not be opened.

*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to load member information
Please type 8 to exit the system
Enter your selection Here: 7

***Load member information***
Please enter the member file name: 3_1.txt
The file 3_1.txt has been loaded...
```

3_1.txt

Name	Password
Andy	12345678
Kevin	227788
Jinx	456456
Akali	77777
Jax	878787

Lab 12.3

NCU Theater Reservation System(IV)

◆The screen dialog should appear as follows:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to load member information
Please type 8 to exit the system
Enter your selection Here: 6

***NCU Theater Membership***
No.  Username
1.   Andy
2.   Kevin
3.   Jinx
4.   Akali
5.   Jax
```

3_1.txt

Name	Password
Andy	12345678
Kevin	227788
Jinx	456456
Akali	77777
Jax	878787

Lab 12.3

NCU Theater Reservation System(IV)

◆The screen dialog should appear as follows:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to load member information
Please type 8 to exit the system
Enter your selection Here: 5

***NCU Theater Member Center***
Do you have the membership? (y/n) y
Enter Username: Jinx
Enter Password: 456456

*** Welcome to NCU theater reservation system ***
***Welcome back Jinx***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to log-out
Please type 6 to show member
Please type 7 to load member information
Please type 8 to exit the system
Enter your selection Here: 5

*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to load member information
Please type 8 to exit the system
Enter your selection Here:
```

3_1.txt

Name	Password
Andy	12345678
Kevin	227788
Jinx	456456
Akali	77777
Jax	878787

Lab 12.3

NCU Theater Reservation System(IV)

◆思考方向：

使用功能7 要求user輸入讀取file檔名 錯誤則返回menu → 讀取input file將會員資料存入陣列或vector中 → 使用功能6 檢查會員資料是否成功載入

Lab 12.3
NCU Theater Reservation System(IV)

◆Step. 1

```
void load_member()
{
    int i = 0, password;
    string name;
    string infilename;

    cout << "***Load member information***" << endl;
    cout << "Please enter the member file name: ";
    cin >> infilename;

    infile.open(infilename, ios::in);

    if(!infile)
    {
        cout<<"File could not be opened. "<< endl << endl;
        return;
    }
    else                    使用seekg讓程式不要讀到檔案第一行內容
    {
        infile.seekg(19,ios::beg);
        while (infile >> name >> password)
        {
            member_name[i] = name;
            member_data[i][0] = password;
            i++;
            member_number++;
        }
    }
    cout << "The file " << infilename << " has been loaded...\n"<< endl;

    return;
}
```

將會員資料讀進程式

若前面的資料格式有建好
這題在執行上就會非常簡單

Lab 12.4
Find A Word

- In this problem, you need to write a program which can find a specific word from a given file **“4_1.txt”.**
- In the beginning, the program will read the file, take a word as input, and then start to find the word you entered.
- If the string could be found in the 4_1.txt, the program will show the message **“True”.**
- If not, the program will show the message **“False”,** and the program will continue to take the next word as input and repeat its execution until the user enters -1.

Lab 12.4
Find A Word

- Requirement:
  - You must to use the instruction below to open the file, otherwise you will get 0 point.
    - **infile.open("4_1.txt", ios::in);**
  - Note that the word can be found in the vertical and horizontal in any of the directions. Moreover, the word can be found as combinations of vertical and horizontal series in any direction.

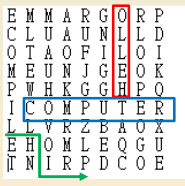
Lab 12.4
Find A Word

- The screen dialog should appear as follows:

```
HELLO
True
COMPUTER
True
LLVOIRP
True
ZZZZZ
False
-1

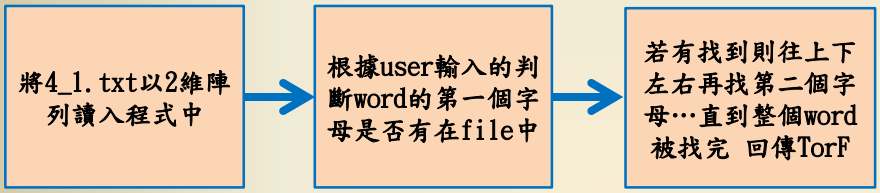
Process returned 0 (0x0)   execution time : 143.415 s
Press any key to continue.
```

4_1.txt



Lab 12.4
Find A Word

◆思考方向：



Lab 12.4
Find A Word

◆Step. 1

```
int r=0, c=0;
char letter;
string word;
vector<vector<char>> board(9, vector<char>(9));

infile.open("4_1.txt", ios::in);

while(infile >> letter)
{
    board[r][c]=letter;
    if(c < 8) c++;
    else
    {
        c=0;
        r++;
        if(r>=9)    break;
    }
}

infile.close();
```

因題目的file固定為9*9的大寫字母 能以此方式將字母存入2維陣列board中

Lab 12.4
Find A Word

◆Step. 2

```
bool exist(vector<vector<char>>& board, int i, int j, string& word, int idx)
{
    if(i >= board.size() || j>= board[0].size() || board[i][j] != word[idx])  return false;
    if(idx == word.size()-1) return true;

    char temp = board[i][j];
    board[i][j] = '*';

    if(exist(board, i+1, j, word, idx+1)) return true;
    if(exist(board, i-1, j, word, idx+1)) return true;
    if(exist(board, i, j+1, word, idx+1)) return true;
    if(exist(board, i, j-1, word, idx+1)) return true;

    board[i][j] = temp;
    return false;
}
bool search(vector<vector<char>>& board, string word)
{
    if(board.empty()||word.empty()) return false;

    for(int i = 0; i<board.size(); i++)
        for(int j = 0; j<board[0].size(); j++)
            if(exist(board,i,j,word,0)) return true;

    return false;
}
```

根據search丟進來的位置用遞迴的方式 上下左右尋找word 找到則回傳True

用兩層迴圈尋找board中輸入之word的第一個字母 丟進exist的function中

Lab 12.4
Find A Word

◆Step. 3　整串找完則回傳T 否則回傳F

```
while(getline(cin,word))
{
    if(word == "-1") return 0;
    else
    {
        if(search(board, word)) cout <<"True"<< endl;
        else cout << "False" << endl;
    }

}
```



Evaluation



- Lab12.1(30%)
  - 5 testcase, 6% each.
- Lab12.2(30%)
  - 5 testcase, 6% each.
- Lab12.3(30%)
  - 5 testcase, 6% each.
- Lab12.4(20%)
  - 5 testcase, 4% each.

